

MCS 320 Project Three : Modeling Competing Species due Monday 3 August 2026, at 2pm.

The goal of the project is to explore a computational model of competing species. SageMath is used to solve differential equations, to plot the solution trajectories, and then to interpret the model.

1. A Population Model

Let $P(t)$ denote the size of a population at time t . Then the evolution of $P(t)$ as function on time can be described via the initial value problem

$$\frac{d}{dt}P(t) = r_p P(t), \quad P(0) = P_0,$$

where r_p is the growth rate. The differential equation expresses that the change in the population is directly proportional to its size. As this leads to exponential growth, this model is not realistic as growth is limited by the carrying capacity K_p , used in the differential equation

$$\frac{d}{dt}P(t) = r_p \left(1 - \frac{P(t)}{K_p}\right) P(t).$$

As $P(t)$ grows closer to its carrying capacity K_p , the factor $1 - P(t)/K_p$ ensures a slowing growth.

The growth of P is further restricted if there is another species Q competing for the same resources needed by P . The coefficient c_p measures how much P is influenced by Q :

$$\frac{d}{dt}P(t) = r_p \left(1 - \frac{P(t)}{K_p} - c_p Q(t)\right) P(t), \quad \frac{d}{dt}Q(t) = r_q \left(1 - \frac{Q(t)}{K_q} - c_q P(t)\right) Q(t).$$

We thus have a system of two nonlinear differential equations to model the growth of $P(t)$ and $Q(t)$, with growth rates r_p, r_q , carrying capacities K_p, K_q , coefficients c_p, c_q , and initial values P_0, Q_0 .

Assignment One. At an equilibrium, the sizes of the populations do not change, or equivalently:

$$\frac{d}{dt}P(t) = 0, \quad \frac{d}{dt}Q(t) = 0.$$

To compute all equilibria, solve the polynomial system defined by the equations

$$r_p \left(1 - \frac{P}{K_p} - c_p Q\right) P = 0, \quad r_q \left(1 - \frac{Q}{K_q} - c_q P\right) Q = 0,$$

for P and Q symbolically, by bringing the system into triangular form. For symbolic parameters $r_p, r_q, K_p, K_q, c_p, c_q$, how many equilibria for (P, Q) can we have?

2. Computing Trajectories

To prepare for a numeric solution, the above system is rewritten in a more streamlined representation:

$$\frac{d}{dt}P(t) = \left(r_p - \alpha_p P(t) - \beta_p Q(t)\right) P(t), \quad \frac{d}{dt}Q(t) = \left(r_q - \alpha_q Q(t) - \beta_q P(t)\right) Q(t),$$

where for the parameters we use numerical values $r_p = 1, \alpha_p = 1, \beta_p = 1, r_q = 3/4, \alpha_q = 1, \beta_q = 1/2$.

Assignment Two. Setup the numerical solving of the initial value problem, selecting at least four different pairs of random values for the initial values (P_0, Q_0) inside $[0, 1]$.

1. Compute the solution trajectories for a long enough time span, till an equilibrium is reached.
2. Make a plot of at least four solution trajectories, starting at clearly different start values (P_0, Q_0) . On the plot, mark all numerical equilibria with big red dots.
3. Construct an animation of the trajectories, using at least 10 frames for increasing values of time, for all four trajectories.

Write at least one paragraph summarizing your observations of the computational experiments.

3. Attracting and Repelling Equilibria

Some equilibria are attracting: all trajectories converge to an attracting equilibrium. Other equilibria are repelling: even as we start very close to a repelling equilibrium, the trajectory moves away from it.

Assignment Three. Which of the equilibria in this numerical model are repelling and attracting? Justify your answer by making systematic plots of solution trajectories.

4. The Deadline is Monday 3 August, at 2pm.

Upload your answer to gradescope at the latest on Monday 3 August, before 2pm.

The solution consists of one single notebook, organized according to the assignments. Mark the start of each solution to an assignment using a heading in a cell. Your notebook should run from top to bottom as a program without errors. Apply proper formatting in your notebook so it reads like a technical report if you would print it. Document the execution cells with complete sentences, properly formatted in markdown cells.

You may (not must) work in pairs for this project. A pair consists of two, not three or more. If you decide to work in a pair, then you must send me an email with the name of your partner and with the email address of your partner in the copy of the email, before 2pm on Wednesday 29 July. If working in a pair, then only one Jupyter notebook should be submitted.

If you have questions, concerns, or difficulties, feel free to contact me for help.

References

- [1] William E. Boyce, Richard C. DiPrima, Douglas B. Meade. *Elementary Differential Equations and Boundary Value Problems*. 11th Edition, Wiley, 2017.